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Two high-altitude balloon flights: payload design, flight data, and a reaction-wheel experiment

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30,849 m
Flight 2 peak (GPS), 101,211 ft

31,761 m
Flight 1 peak (GPS), 104,203 ft

12,245
Flight 2 science records, about 1 Hz

143.9 km
Flight 2, launch to last GPS fix

1 Summary

I launched a weather-balloon payload three times in 2026. On the first attempt, the swivel connecting the parachute to the payload broke when I released the balloon a second time, and the balloon and parachute flew away without the payload. After I redesigned the flight train, both later flights climbed past 30 km and I recovered both payloads. Flight 1 reached 31,761 m (104,203 ft) with a GPS logger and a camera. Flight 2 carried ten environmental sensors, a two-Arduino logger, three cameras, and a single-axis reaction wheel, and reached 30,849.2 m (101,211 ft). The flight-log numbers in this report come from the raw files in the repository, and the scripts in analysis/ regenerate every figure and calculated value. Bench values, camera timings, and my own observations are labeled as such.

The main results from Flight 2 are a near-continuous pressure, temperature, light, and acceleration record at about 1 Hz (12,245 records, 10,303 of them after release), a GPS track with one large gap, and a clear picture of which parts of the payload worked and which did not. The reaction wheel launched running and worked in a post-flight ground test, but nothing recorded it in flight, so its in-flight performance is unknown.

	Failed launch	Flight 1	Flight 2
Date	Before Aug 2026	Aug 7, 2026	Oct 28, 2026
Site	Mojave Desert, CA	Lucerne Dry Lake, CA	Stoddard Valley OHV Area near Barstow, CA
Balloon	Latex weather balloon	Latex weather balloon	Kaymont TX1200
Payload	XPS foam box: electronics, tracker, camera	XPS foam box: GPS logger, SPOT Trace, action camera	XPS foam box: 10 sensors, Mega + Uno logger, GPS, SPOT Trace, Mobius Maxi 4K, 2 RunCam Thumb 2, reaction wheel; box about 3.89 lb before external hardware
Peak altitude (GPS)	Not measured (a few meters)	31,761 m (104,203 ft) at 23:57:05 UTC	30,849.2 m (101,211 ft) at 22:08:56 UTC
Outcome	Swivel broke on the second release; balloon and parachute lost	Recovered; 158.9 km (98.7 mi) from launch	Recovered; 143.9 km (89.4 mi) from launch, about 32.5 km (20.2 mi) east of Baker, CA
What changed next	Stronger swivels, two lines, and extra lines through the foam body	Added the science logger and the reaction wheel	Next: log the wheel, add a swivel or more wheel authority, fix the GPS gap and the camera dropout (section 8)

Table 1 Mission summary. Altitudes are GPS altitude above mean sea level.

2 The failed launch and the flight-train redesign

The first launch failed because of the one part I had treated as simple: the connection between the parachute and the payload. I had spent my time on the electronics and the tracking, and a single swivel joined the parachute line to the payload.

On the first release the balloon was underfilled, so the payload lifted only a few meters and drifted until I caught it. I added helium and released it again, and as the line pulled tight to lift the payload, the swivel broke. The balloon and

parachute flew away without the payload. There is no flight data from that attempt, only a short video of the payload drifting before I caught it.

The redesign adds redundancy where the first attempt had none and replaces the part that failed. Starting with Flight 1, I used stronger swivels than the one that broke. Two lines now run from the flight train to the payload, and extra lines are routed through the foam body of the box itself, so the added lines reduce the risk of losing the payload to a single attachment failure (Figure 1). This rigging flew on both later flights without a problem, and the recovery photos (Figures 23 and 24) show the bridle lines still in place across the lid after landing. Next time I would load-test the whole flight train with the payload hanging from it before filling the balloon, instead of trusting part ratings on paper. I would also check free lift before release so the balloon is filled right the first time.

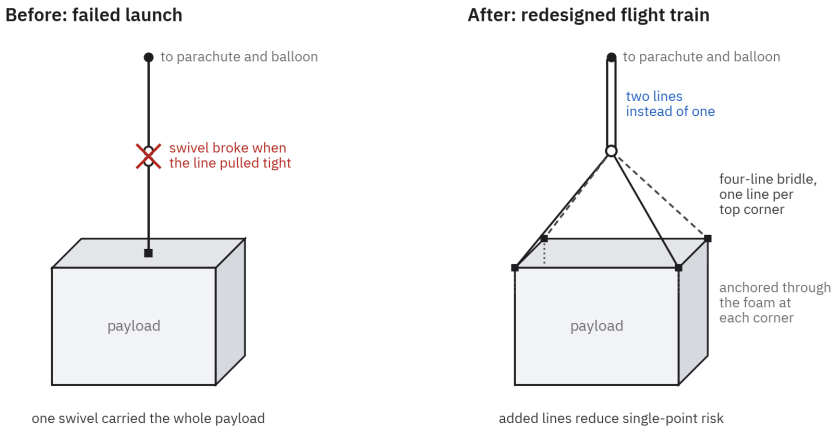


Figure 1 Rigging before and after the failed launch (schematic, not to scale). The after panel shows Flight 2, drawn from my description and the recovery photos (Figures 23 and 24).

3 Payload hardware

The Flight 2 payload is a pink XPS foam box wrapped in reflective insulation and black tape (Figures 3 and 4). The box weighed about 3.89 lb before the external hardware was attached; I did not record the all-up flight mass. Inside are the Arduino Mega 2560, the Arduino Uno with its GPS logger shield, and the PCA9548 multiplexer with the sensor breakouts on STEMMA QT cables. The reaction wheel sits over its motor and controller board. Two Nitecore USB power banks (one runs the wheel's Nano), a LiPo battery that drives the wheel's motor through the G2 driver, and two battery holders supply power. The cameras are a Mobius Maxi 4K and two RunCam Thumb 2s. The balloon was a Kaymont TX1200. The parachute model and line specifications were not recorded.

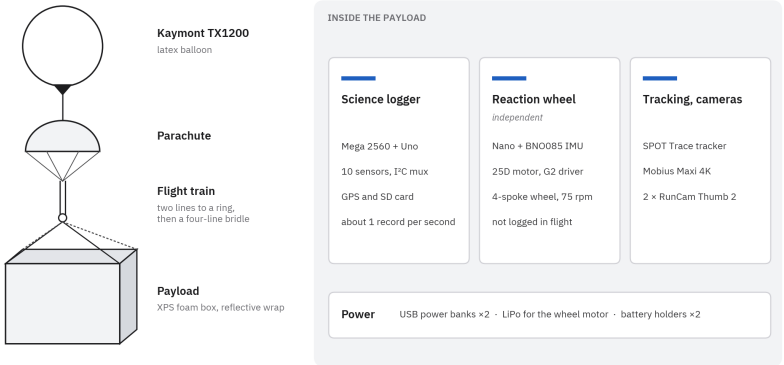


Figure 2 Flight 2 system block diagram. Source: my build notes and the Flight 2 firmware.

Sensor placement comes from the mux map in the Mega firmware: the bottom exterior cluster holds MCP9808 #3, TMP119, MS8607, AHT20, and BMP390; the VEML7700 and LTR390 face up; MCP9808 #1 measures interior air; MCP9808 #2 sits in the battery area; and the LIS3DH accelerometer is on the rigid interior deck. The exterior cluster is not shielded from the sun or ventilated, which matters for the temperature results in section 5.

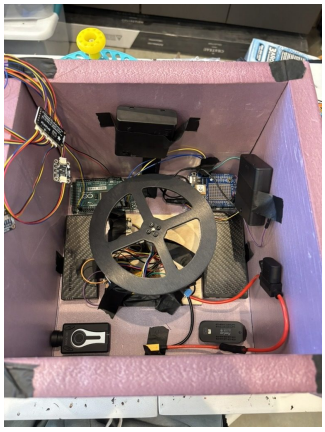


Figure 3 Flight 2 payload with the lid off before launch: Mega (left), Uno with GPS shield (right), mux and TMP119 (upper left), reaction wheel (center), two power banks, Mobius (bottom left), RunCam (bottom right).



Figure 4 Flight 2 payload closed and wrapped, at the Stoddard Valley launch site during balloon fill, Oct 28, 2026.

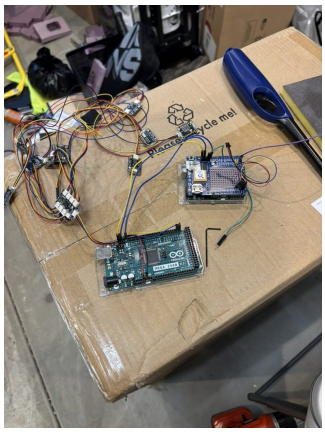


Figure 5 Science electronics on the bench before installation: Mega 2560 R3, Uno with Adafruit GPS logger shield, PCA9548 mux, and sensor breakouts.



Figure 6 Reaction wheel installed in the payload (top view): 4-spoke wheel over the motor, driver, and Nano.

4 Avionics and data flow

The Mega sweeps the sensors once a second through the multiplexer, builds one record, and sends it to the Uno over I2C as a CRC-16 frame in 28-byte chunks with an acknowledgement for each chunk (Figure 7). The Uno writes the record to the SD card as an S row and also writes every checksum-valid GGA and RMC sentence from its own shield GPS as a G row. The Uno uses the RTClib library only for date arithmetic on the GPS time; there is no clock chip on the payload. Firmware versions: Mega 1.2.0-I2C, Uno SCI-UNO-1.3.0-GPS-UNO (printed in the first record of SCI_000.CSV), and the reaction wheel HAB-RW 1.2.0-ground.

The log has 12,245 S rows, 24,822 G rows, 141 E (event) rows, two B (boot) rows, and one D (sensor detection) row. Each S row carries a test flag and three hex bitmasks: valid (sensor read succeeded), oor (value outside the expected range but still logged), and synth (placeholder values from bench-test mode). A failed read leaves the field empty, never zero. Every record in this file has test = 0 and synth = 0, so nothing had to be excluded. The I2C link was not

lossless: the Uno logged 98 SEQ_GAP events totalling 520 missing Mega sequence numbers out of 12,765 generated, about 4.1%.

One thing the log does not contain is a GPS fix from the Mega. The firmware reserves gps_* columns for a receiver on the Mega's Serial2, but gps_fix is 0 in every one of the 12,245 records and the other gps_* fields are blank. All GPS positions in the science log come from the Uno shield's receiver in the G rows; SPOT Trace positions come from the KML and are marked separately. The reaction wheel is a separate system: its Nano streams 10 Hz telemetry only over USB serial, which nothing recorded in flight, and keeps an EEPROM summary of only its most recent armed run.

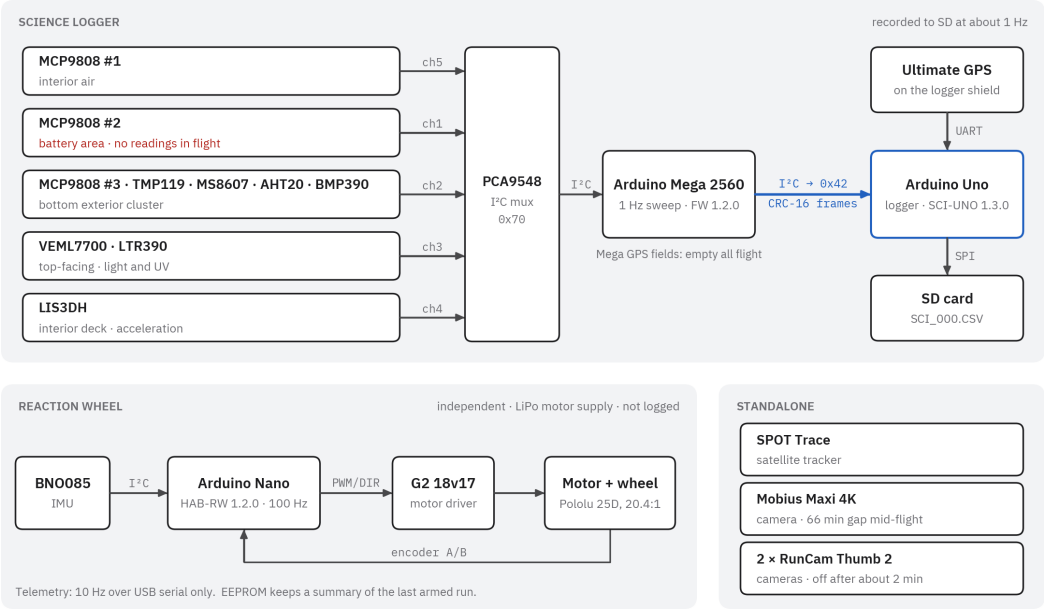


Figure 7 Flight 2 avionics and data flow. Source: HAB_Mega_Science.ino, HAB_Uno_Logger.ino, sci_link.h, HAB_RW_Controller.ino.

5 Flight 2 results

5a Altitude, release, and burst

Release was at 19:42:55 UTC. The accelerometer recorded a 2.53 g jolt at that second, and GPS altitude starts climbing from 894 m three seconds later. The highest GPS fix was 30,849.2 m (101,211 ft) at 22:08:56 UTC by the receiver's own timestamp; the next fix is lower, so burst was at about that time. Two other sensors agree: the MS8607 reached its minimum of 10.14 hPa at 22:08:57 UTC (logger clock), and starting at 22:08:55 UTC (logger clock) 4 of the next 5 accelerometer samples read below 0.2 g, with a minimum of 0.028 g (the other read about 0.6 g). That is consistent with free fall, but sampling once a second cannot show how long it lasted. The ascent took 146.0 min at a mean 3.42 m/s.

The last GPS fix was at 22:43:20 UTC at 35.2834 N, 115.7167 W, 1,794 m, with 20 satellites; GPS heights this late in the descent are suspect. A SPOT Trace report at 22:44:03 UTC, 57 m from that fix at 35.28354 N, 115.71609 W, still lists 10.3 mph; the next report, at 22:49:03 UTC, lists 0.0 mph at 1,238 m, and USGS elevation data put the ground there at about 1,245 m. The payload landed between about 22:43 and 22:49 UTC; I cannot pin the exact time. The straight-line distance from launch to the last GPS fix is 143.9 km (89.4 mi), and the path through the retained fixes, with straight lines across the gaps, is 152.8 km, a lower bound. I excluded GPS fixes before 19:20 UTC: the receiver had 2 to 9 satellites then and reported altitudes between 188 and 964 m while sitting on the ground.

The GPS lost its fix for 52.3 min on the way up, from 20:33:46 UTC at 10,486 m to 21:26:04 UTC at 21,006 m, and three more times on descent (5.2, 3.1, and 10.1 min). The receiver kept sending no-fix sentences the whole time, so it was alive but not tracking. I do not know why. The same receiver family worked to 31,761 m on Flight 1 and to

30,849 m later on this flight, so an altitude limit does not explain it. Where GPS is missing, I use pressure altitude from the MS8607 through the 1976 U.S. Standard Atmosphere, and I label every such value as an estimate.

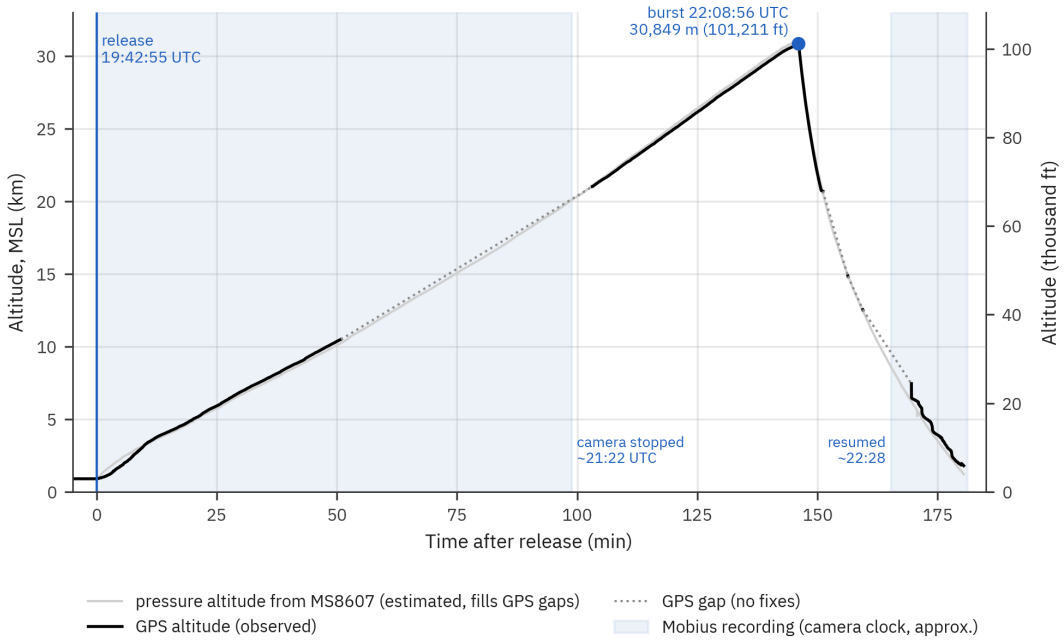


Figure 8 Flight 2 GPS altitude versus time with release, burst, GPS gaps, and the Mobius recording window. Camera times are approximate (camera clock about 15 min 7 s fast, within about 1 min). Source: SCI_000.CSV.

5b Ascent and descent rate

The ascent rate was below the 5.0 m/s I had assumed in the prediction for almost the whole climb: about 2.8 to 3.2 m/s between 4 and 10 km and 3.8 to 4.0 m/s between 21 and 30 km (Figure 9). The descent started at about 42.8 m/s above 25 km, where the low air density makes the descent much faster, and settled to 7.6 m/s between 7 and 3 km. GPS heights near the end of the descent are suspect, so I do not use them to judge the landing.

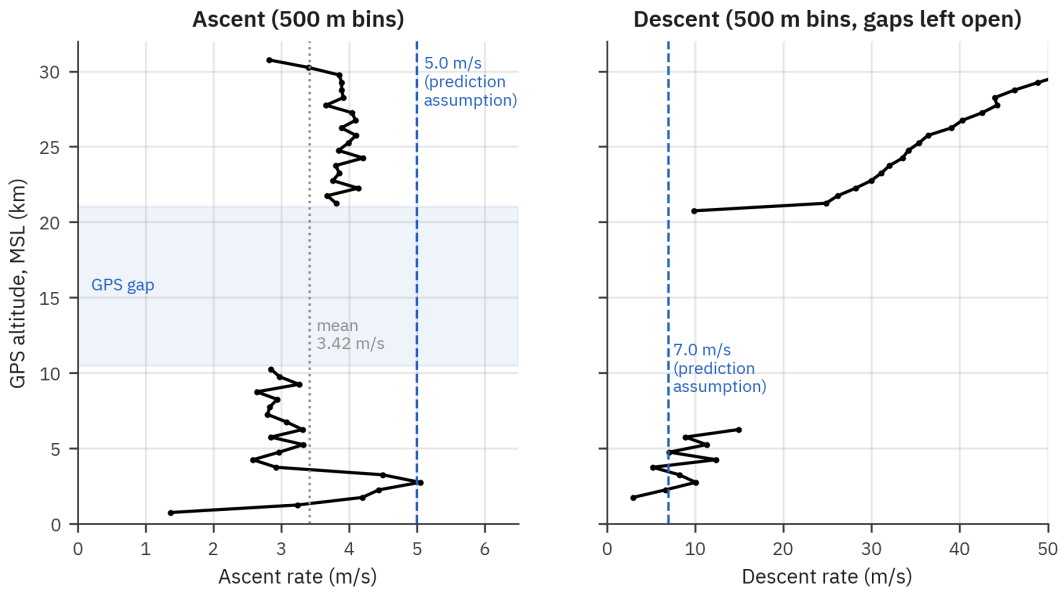


Figure 9 Ascent and descent rate from GPS altitude in 500 m bins. Lines are broken where the GPS had no fix. Source: SCI_000.CSV G rows.

5c Pressure against the standard atmosphere

On the ascent, above 2 km, with GPS height converted to geopotential height, the MS8607 tracked the 1976 U.S. Standard Atmosphere within -4% to +4% wherever GPS altitude was available (Figure 10), from 912 hPa on the ground to 10.14 hPa at burst. The firmware marks anything below 10 hPa as outside the sensor's normal range, so the last readings near burst are at the edge of what the part is rated for. The BMP390 followed the MS8607 closely until 21:03:55 UTC at 100 hPa and then stopped tracking: it read between 103 and 118 hPa for the rest of the ascent while the MS8607 fell to 10 hPa. Its qualified range is 300 to 1250 hPa in both the firmware and the datasheet, so the divergence below about 100 hPa is consistent with operating outside its qualified range; the firmware flagged 6,846 of its readings as out of range.

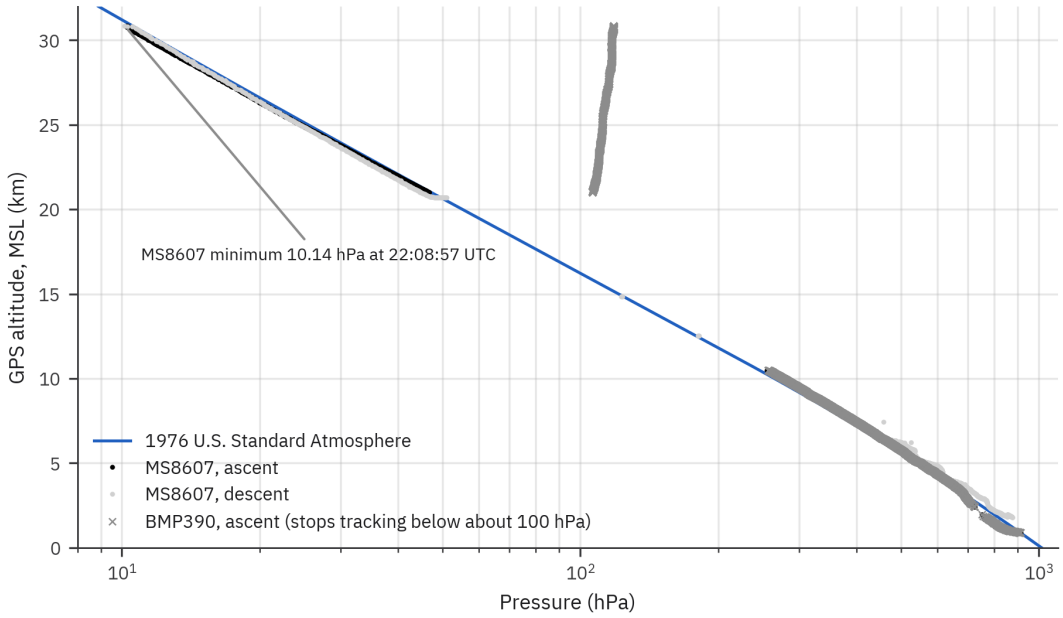


Figure 10 MS8607 and BMP390 pressure versus GPS altitude with the 1976 U.S. Standard Atmosphere. Source: SCI_000.CSV.

5d Temperature

The coldest reading of the flight was -58.31 °C on MCP9808 #3 at 22:22:38 UTC on descent, at 187 hPa (pressure altitude about 12.2 km). That is below the MCP9808's rated range, which ends at −40 °C, so its 846 readings colder than −40 °C are outside the part's specification. The TMP119 and MS8607 minima were -56.9 and -53.4 °C. On ascent the exterior minimum was only -38.94 °C at 167 hPa (about 12.9 km), and above that the exterior sensors warmed steadily to between 3.1 and 11.9 °C near burst, where the standard atmosphere is -46.6 °C (Figure 11). I do not believe those high-altitude readings are air temperature. The cluster hangs under the box in full sun with no shield and no forced ventilation, and at a few hectopascals there is almost no convection to carry away absorbed sunlight. The descent profile, taken at 8 to 40 m/s with much more airflow, reads much colder at the same pressure altitude; these are uncalibrated sensor readings, not air temperatures. The cold layer is visible in both profiles between about 170 and 190 hPa, which is where I would expect the tropopause.

Interior air (MCP9808 #1) was 37.2 °C at release after the box sat in the sun, fell to about 10 °C near burst, and reached its minimum of -18.69 °C at 22:29:02 UTC on descent. The foam and the electronics' own heat kept the inside above the outside by 20 to 40 °C through the coldest part of the flight. MCP9808 #2 in the battery area returned no readings at all (section 6), so I have no battery temperature.

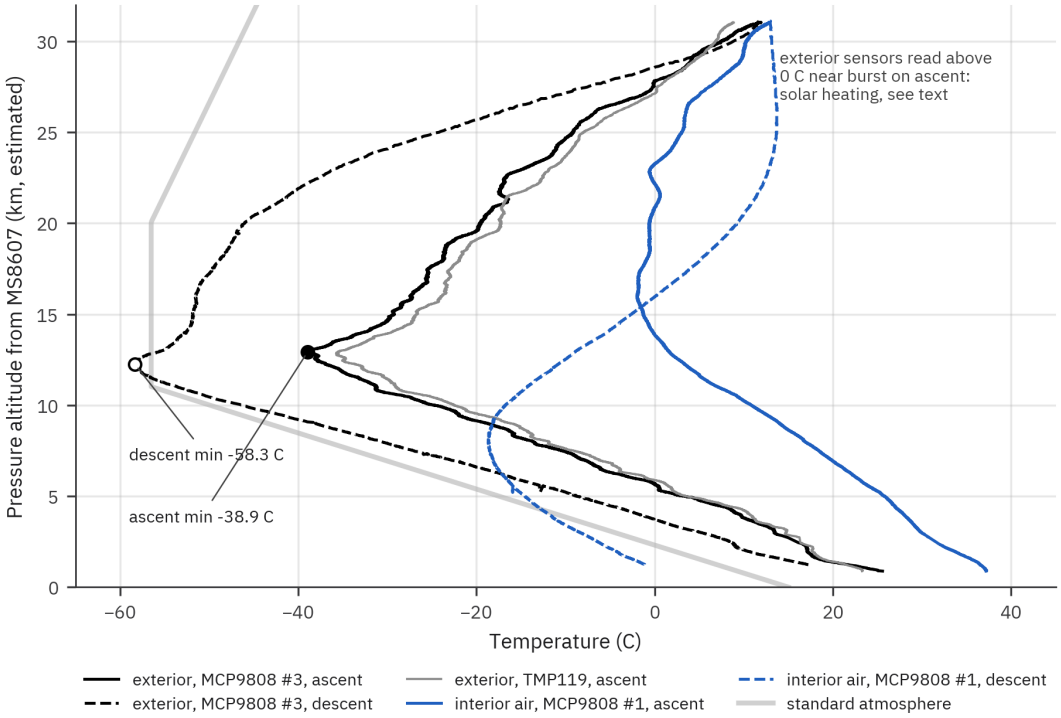


Figure 11 Exterior and interior temperature versus pressure altitude from the MS8607 (an estimate, used because GPS was missing between 10.5 and 21 km on ascent). Source: SCI_000.CSV.

5e Horizontal wind from GPS

Treating the payload's horizontal speed over the ground as the wind, the balloon drifted at about 12.1 m/s between 5 and 10 km, slowed to about 7.8 m/s near 23 to 24 km, and then sped up again to a median of 24.2 m/s in the 30 km bin just below burst (Figure 12). The GPS gap means there is no wind data between 10.5 and 21 km. The track ran east-northeast the whole flight (Figure 13).

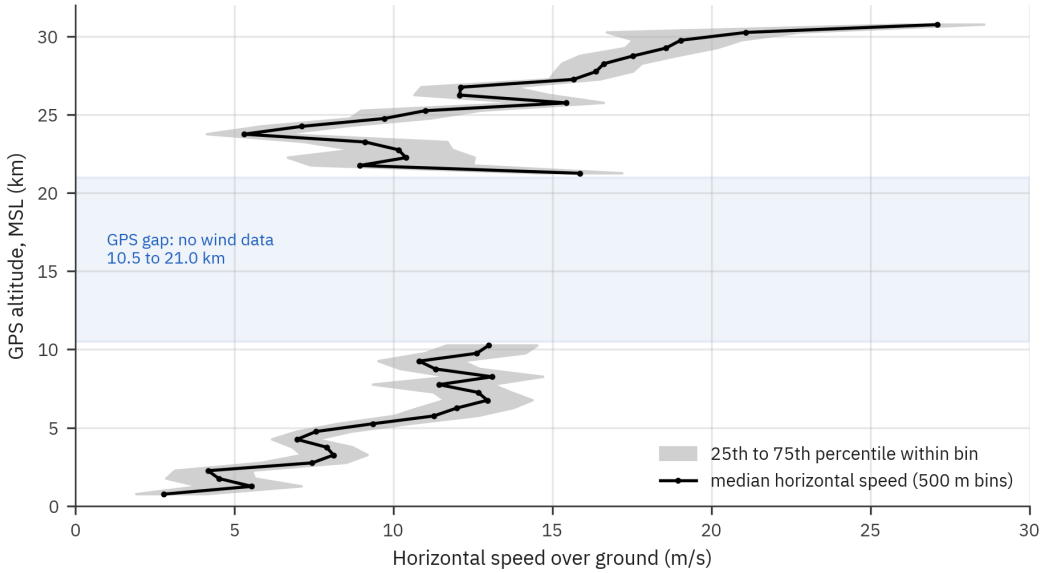


Figure 12 Horizontal speed over the ground from consecutive GPS fixes during ascent, 500 m bins. Source: SCI_000.CSV G rows.

5f Ground tracks

Flight 1 flew west from Lucerne Dry Lake and landed 158.9 km (98.7 mi) away; Flight 2 flew east-northeast from Stoddard Valley and landed 143.9 km (89.4 mi) away. Both tracks have gaps. On the Flight 2 map I show the SPOT Trace fixes as a separate marker: 11 of them fall inside the 52.3-min GPS gap, which is why the gap's straight connector is a fair stand-in for the real path. The SPOT export is stamped in UTC minus 5 hours; I checked the offset against release.

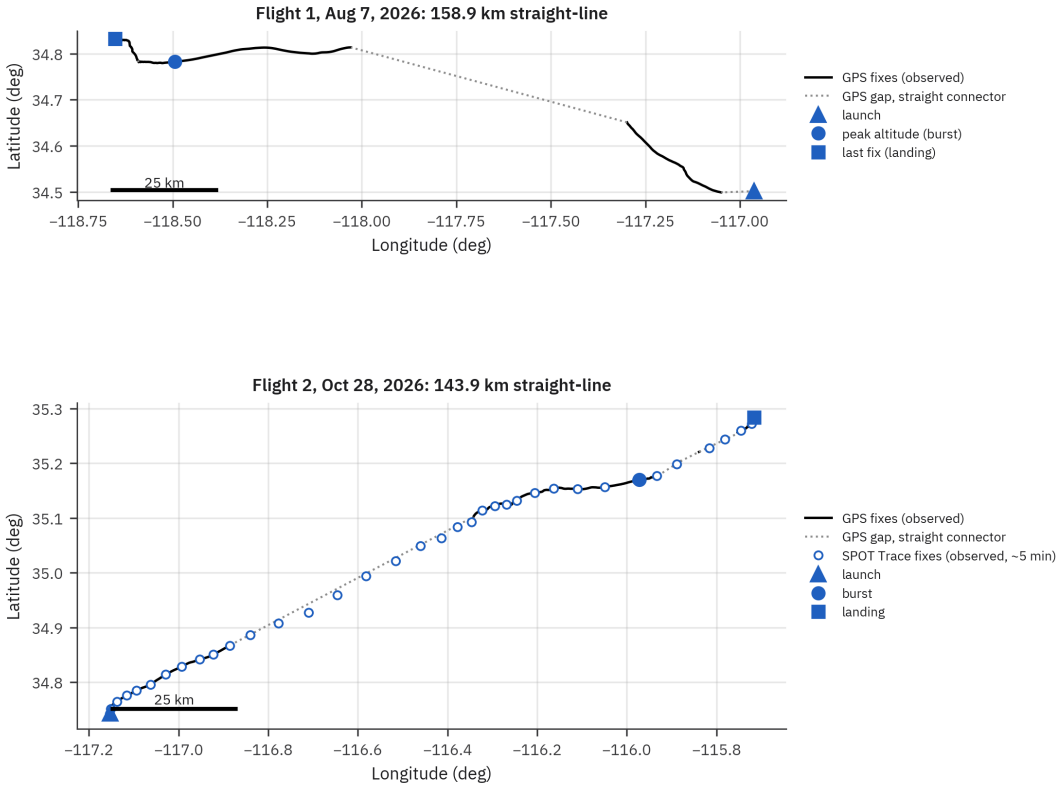


Figure 13 Ground tracks of both flights. Observed GPS fixes are solid black, gaps are dotted connectors, SPOT fixes are open circles. Source: Aug_7_2026_GPS_Fixes.csv, SCI_000.CSV, Second_Successful_Launch.kml (SPOT folder).

5g Prediction against reality

The prediction run I used, which assumed a 19:44 UTC launch from the site, also assumed a 5.0 m/s ascent, a 7.0 m/s descent, and burst at 32,000 m. It put burst at 35.0453 N, 116.3044 W at 21:27:36 UTC and landing at 35.14775 N, 116.09071 W at 22:00:15 UTC. The real burst was 33.2 km from the predicted one and the SPOT landing position was 37.2 km from the predicted landing, and the predicted landing time was before the real burst.

The slower ascent is a plausible major contributor to the miss, though I have not separated it from forecast winds, timing, and burst altitude. The prediction had the balloon aloft for 103.6 min on the way up; it actually took 146.0 min at 3.42 m/s, so it rode the wind for 42 extra minutes and drifted 117.5 km from launch to burst instead of the predicted 84.3 km. The descent legs were close: 22.5 km in 32.7 min predicted, 26.4 km in 35.1 min from the highest fix to the first SPOT report near the landing site (touchdown time is uncertain). The predicted burst altitude was also 1,150 m too high. The KML file holds 13 prediction runs; only the 19:44 run is used here, and its landing placemark carries a ground altitude of 284.8 m that was copied from the prediction, not measured; the ground at the landing site is about 1,245 m (USGS).

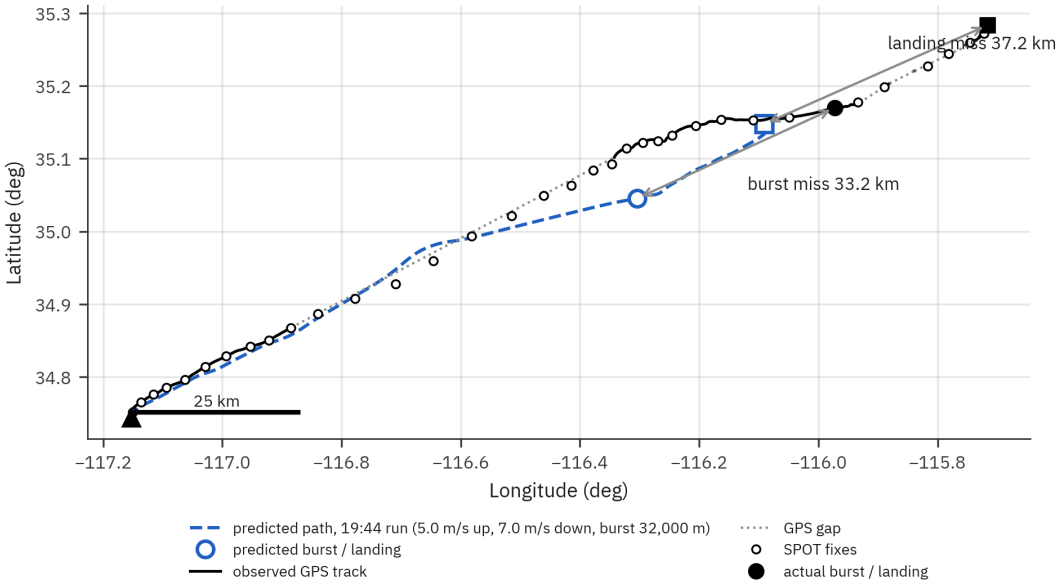


Figure 14 Predicted path (dashed) against the observed GPS track and SPOT fixes for Flight 2. Source: Second_Successful_Launch.kml (Prediction folder) and SCI_000.CSV.

5h Light and UV

Both top-facing sensors worked for the whole flight and neither saturated (VEML7700 raw maximum 17,483 of 65,535). The VEML7700 read about 25.8 klux below 2 km and 21.9 klux near burst; the LTR390 UV count rose from about 614 below 2 km to 820 near burst (Figure 15). Both sensors turned with the payload and their view of the sun changed as it did, so I read these as usable but qualitative: the UV increase with altitude is the expected direction, and the illuminance is nearly flat.

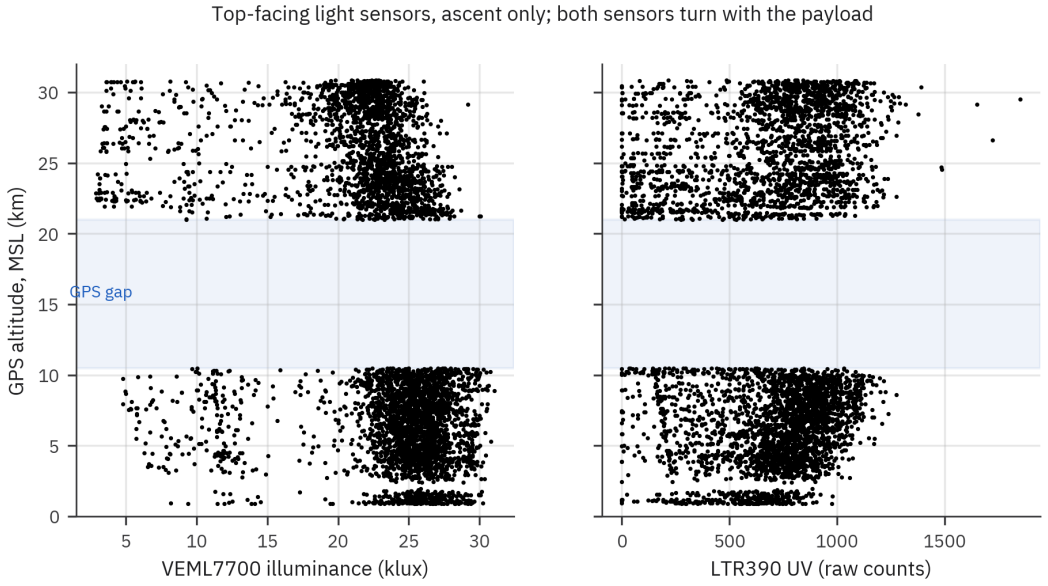


Figure 15 VEML7700 illuminance and LTR390 UV counts versus GPS altitude on ascent. Source: SCI_000.CSV.

5i Acceleration

After the release jolt, the acceleration magnitude on ascent never exceeded 1.22 g (one sample at 19:43:29 UTC, 34 s after release); the 99th percentile was 1.11 g and only 1 sample was above 1.17 g. The one large event is burst: low-g readings consistent with free fall, then a cluster of 1.5 to 2.23 g readings from 22:09:43 to 22:11:46 UTC as the parachute and the falling balloon remains loaded the box (Figure 16). The accelerometer showed no large acceleration around 21:21:32 UTC, when the Mobius footage shakes violently just before it stops: the maximum in that three-minute window is 1.08 g. However, its 1 Hz sample rate cannot rule out brief vibration or rotation.

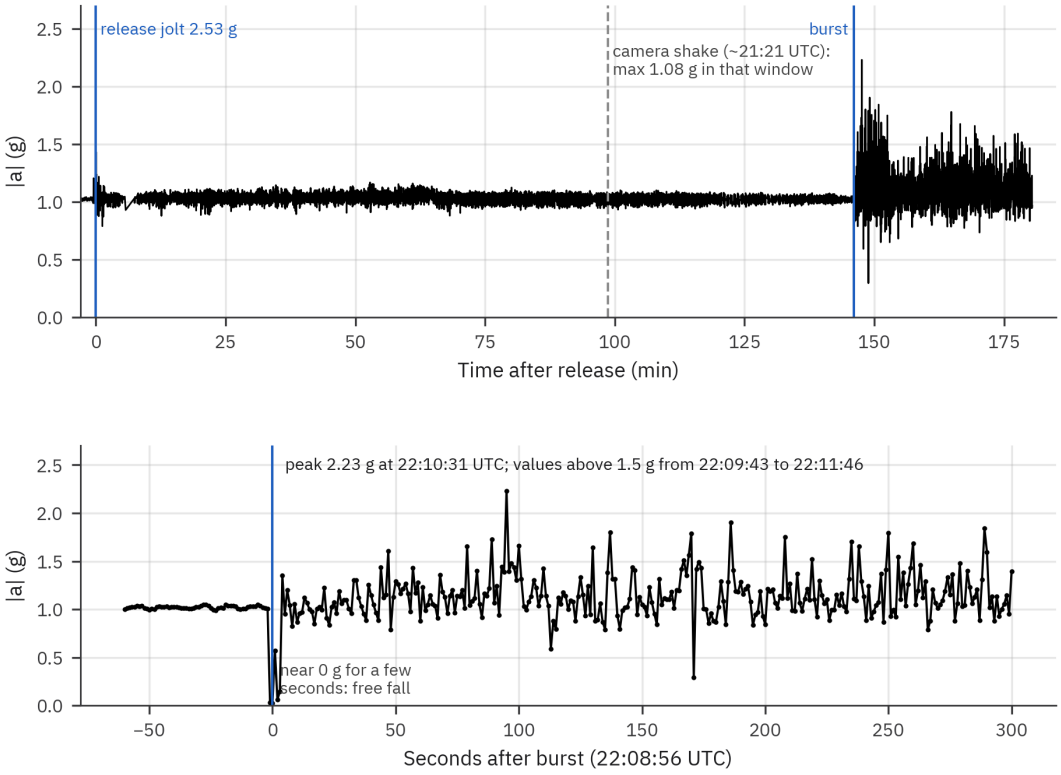


Figure 16 Acceleration magnitude from the LIS3DH for the whole flight (top) and around burst (bottom). Source: SCI_000.CSV.

6 Sensor and camera reliability audit

Of the ten sensors, one failed completely, the BMP390 diverged once it left its qualified range, and the two humidity channels produced nothing usable. The rest returned a reading in every one of the 12,245 records, though several logged values outside their rated ranges (Table 2). Two of the three cameras shut down about two minutes after launch, and the third missed burst. The GPS worked but lost its fix for 71 of the 180 minutes from release to the last GPS fix, about 39% of the flight. The details are in Table 2.

Item	Status	Evidence	Likely cause	What I will change
MCP9808 #1 (interior air)	Worked	12,245/12,245 valid; 37 to −19 °C		
MCP9808 #2 (battery area)	Failed	0 valid readings; valid bit never set, column empty	Unknown: wiring, its mux channel, or the part itself	Bench-verify every mux channel in the flight box before closing the lid; log the detect mask on every boot
MCP9808 #3 (exterior)	Worked	12,245 valid; min -58.31 °C; 846 readings flagged below −40 °C spec	Readings below the part's rated range are flagged, not dropped	Shield from sun; aspirate

Item	Status	Evidence	Likely cause	What I will change
TMP119 (exterior)	Worked	12,245 valid; min -56.9 °C; 111 flagged out of range		Same
MS8607 pressure/temp	Worked	912 to 10.14 hPa; on ascent, within -4 to +4% of standard atmosphere (not a calibration)	Below 10 hPa is outside the part's normal range	Keep as primary barometer
MS8607 humidity	Unusable	4,765 negative readings on ascent; 5,749 flagged out of range	Not isolated: cold, low pressure, and sensor response are all possible	Drop or replace with a sensor rated for the stratosphere
AHT20 temp/humidity	Partial	Temperature tracked the others; humidity read 0.0 for 90% of the flight (ground median 23.9%); 821 flagged out of range	Not isolated; cold is one possibility	Treat as temperature only
BMP390	Range limit	Tracked MS8607 to 100 hPa, then stayed at 103 to 118 hPa; minimum 91.0 hPa	Qualified range 300 to 1250 hPa	Keep for the lower atmosphere only
VEML7700 / LTR390	Worked	12,245 valid each; no saturation		Add a fixed-orientation reference or a sun shield
LIS3DH accelerometer	Worked	12,245 valid; release and burst both visible		Raise sample rate for impact events
Uno shield GPS	Partial	7,711 fixes used; 52.3 min gap on ascent (10.5 to 21.0 km) plus 18 min of descent gaps	Unknown; receiver stayed alive but reported no fix	Second receiver on the Mega (its columns were empty); log satellite count and fix status continuously
Mega GPS columns	No data	gps_fix = 0 in all 12,245 S rows; other gps_* fields blank	No receiver delivering fixes to the Mega's Serial2	Connect and verify before flight
Mega to Uno I2C link	Partial	98 SEQ_GAP events, 520 records lost (4.1%)	Chunk retries exhausted or Uno busy writing SD	Buffer and retry on the Mega; log link statistics
Mobius Maxi 4K	Partial	Continuous from before release until about 21:21 UTC (about 20 km pressure altitude), off for 66 min, resumed about 22:28 UTC near 8.6 km; burst not recorded	Possibly the camera itself failing or rattling; the accelerometer shows no large acceleration during the shake, but 1 Hz sampling cannot rule out brief vibration	Insulate and tie down the camera; add a status LED in its view
RunCam Thumb 2 (two)	Failed early	Both stopped about 2 min after launch	Not recorded; cold or power are the candidates	Test at low temperature; power from the main banks
SPOT Trace	Worked	37 fixes in flight at about 5-min intervals; near-landing report 57 m from the last GPS fix (still listing 10.3 mph)		Keep

Table 2 Reliability audit of every sensor and camera on Flight 2. Counts are from SCI_000.CSV; camera times are approximate.

7 Reaction wheel

7a The problem

In the Flight 1 footage the camera's view turned slowly around the horizon for most of the ascent. I saw it; I did not measure it. A balloon payload on a line is free to twist, and the line winds and unwinds as the balloon rotates. I wanted the Flight 2 camera to hold a heading, so I built a single-axis reaction wheel to control yaw.

7b Design and control loop

The controller is an Arduino Nano reading an Adafruit BNO085 IMU and driving a Pololu 25D 20.4:1 geared motor with encoder through a Pololu G2 18v17 driver. The wheel is a custom 4-spoke disk. The motor runs from its own LiPo battery through the G2 driver, and the Nano runs from a USB power bank. It is a bias-momentum design: the wheel idles at 75 rpm and makes torque by speeding up or slowing down inside a 45 to 105 rpm band, never reversing. The control loop runs at 100 Hz, and the telemetry stream at 10 Hz. The firmware latches a fault and stops driving if the payload tilts more than 15 degrees, and the fault stays latched until a serial R command clears it, which nothing could send in flight. The board reported the closed-loop, sign-commissioned configuration: BUILD_COMMISSIONING 0, SIGN_COMMISSIONED 1, HEADING_RATE_SIGN_VERIFIED 1, MDS = 1, SA = 1, BIAS = 1, firmware string HAB-RW 1.2.0-ground, read back from the Nano with the T command after recovery (flight_run_summary.txt). I did not archive the exact binary that flew. The source in the repository is the version I recovered, with its flags set to match that readback. With today's library versions it compiles to 31,220 bytes, about 500 bytes over the standard Nano limit, so the flight build must have differed in a way I can't reconstruct, such as older library versions.

7c Authority calculation

The wheel's moment of inertia is 3.19e-04 kg·m² and the payload's is 0.01926 kg·m² (the wheel's from CAD, the payload's from a bifilar-pendulum test; the original measurement records are not in the repository), so the ratio is 0.0166. The ±30 rpm band around the bias therefore gives only about ±3 deg/s of body-rate authority. If the payload was turning once every 50 s, the unconfirmed video estimate in section 7f (7.2 deg/s), stopping it would need about 72 rpm of wheel speed change, about 2.4 times the band. The balloon keeps twisting the payload through the line during ascent, so a wheel alone will saturate unless the flight train decouples yaw with a swivel or the authority is much larger. If the wheel spun down completely from 75 rpm, its momentum would end up in the payload, about 7.5 deg/s in an isolated-system model. Both inertias are estimates, so these numbers are too; actual yaw and saturation in flight were not measured.

7d What worked

- It flew installed in the Flight 2 payload (Figure 6).
- After recovery the board reported the closed-loop build with its motor and sensor directions checked (status readback).
- At release the wheel was spinning: I saw it and heard it.
- The hardware and firmware survived the flight and the landing intact.
- After recovery, in a 1200 s ground test, the controller's heading error against its own IMU reference peaked at 1.70 deg (Table 3). Absolute heading accuracy was not measured independently.
- Pre-flight bench tests verified the encoder, motor direction, and minimum PWM, and reached about 83 rpm (bench results; the logs are not in the repository).

7e Ground test after recovery

The Nano's EEPROM keeps only its most recent armed run. I re-armed the wheel for a ground test after recovery, so the summary I read back describes that ground test and not the flight. The payload hung indoors from a foam board across a stepladder by a single line, a swivel, and a four-line bridle to the box corners (Figures 17 and 18).

Ground test (not flight)	Value
Run length / time evaluated	1200 s / 1190 s (the 10 s settle after capture is excluded)
Peak heading error	1.70 deg against the IMU's own reference
Time within 2 deg / 3 deg	1188 s / 1188 s, counted only while fully armed and unsaturated
Time LIMITED / saturated	2.0 s near a wheel-speed edge / 0 s
Wheel speed range	59.0 to 97.0 rpm (bias 75, soft band 45 to 105)
How it ended	Fault bit 5, F_TILT: payload tilt over 15 deg

Table 3 Ground-test EEPROM summary (Nano E command), decoded in flight_run_summary.txt.

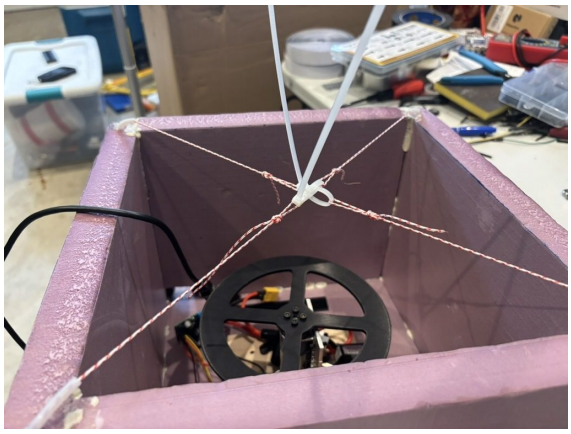


Figure 17 Four-line bridle used for the ground test: lines from the box corners cross to a zip-tie loop above the center.



Figure 18 Post-flight ground test rig: payload hung from a foam board across a stepladder by one line, a swivel, and the four-line bridle.

7f Video evidence (unconfirmed)

I measured payload yaw from the Mobius footage two ways. Frame-to-frame tracking failed: all 44 of 44 windows are flagged as matched poorly, and the heading traces step only when something bright crosses the frame (Figure 20). I do not use the script's steady/swinging/rotating labels, its deg/s, or its swing amplitudes. The second method does not depend on tracking: it asks how long until the view repeats. It found a repeat in 26 windows, 12 of them between 40 and 52 s, spread from 4 to 95 min after the start of the footage (Figure 19). The three strongest repeats were 34.27 s at +60 min, 40.47 s at +95 min, and 50.33 s at +4 min. With the heading drifting one way, a 50 s repeat would mean a full turn about every 50 s, about 7.2 deg/s or 1.2 rotations per minute. I have not verified this by eye, and I present it as suggestive, not as a result. Window times count from camera clock 12:58:04, which is within a few seconds of release under the 15 min 7 s clock offset, with the same ±1 min uncertainty as every other camera time.

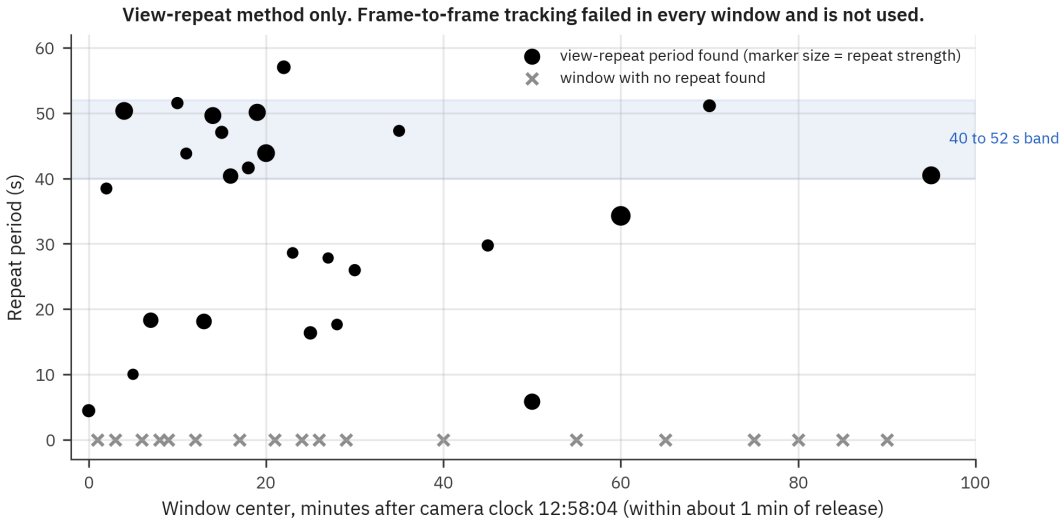


Figure 19 View-repeat period found by the rotation script in each window of the Mobius footage. Windows with no repeat are shown at zero. Source: Flight_2_rotation.csv, Flight_2_upper_rotation.csv.

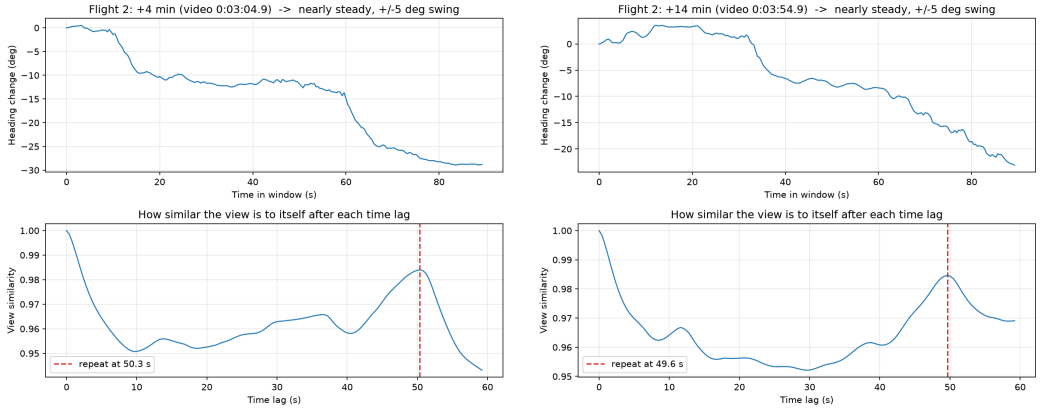


Figure 20 Two windows from the rotation script (+4 min and +14 min) as examples of the frame-to-frame tracker failing: the heading trace (top panels) steps only when bright objects cross the frame, so its deg/s and swing values are not used. The lower panels are the view-repeat similarity curves, which are used. Source: Flight_2__4_min.png, Flight_2__14_min.png.

7g What is unknown, and hypotheses

In-flight heading control is unknown. Nothing recorded the wheel in flight, and I do not know when it stopped. What I know is this: it launched running, it worked in a ground test, and its in-flight performance is unknown. My application described the wheel as partially holding heading during ascent and losing control after burst; the data cannot support either part of that, and this report replaces it. The video suggests, without confirmation, that the payload may have turned about once every 40 to 50 s at times during the first 95 min of ascent.

Possible causes, as hypotheses and not findings:

- A tilt fault at release. The 2.53 g release jolt and the swing that follows could exceed 15 deg; the fault latches until a serial command clears it, which nothing could send in flight.
- A mid-flight reset that left the controller disarmed.
- A power problem, with no battery-voltage log to show it.
- If the estimated rotation rate is correct, the wheel's speed band was about 2.4 times too small to stop it, so even a running wheel could have saturated.

7h Next-flight changes for the wheel

- Log wheel telemetry to SD with timestamps, so the next report can answer the question this one cannot.
- Make faults clear themselves after a few seconds of normal readings instead of latching until a serial command clears them.
- Require tilt past the limit for about 2 s, not one reading, before faulting.
- Re-arm automatically after a reset.
- Log battery voltage.
- Increase authority with a heavier wheel or a wider speed band, or add a swivel so the balloon's twist does not reach the payload.
- Put a status LED in the camera's view.

8 Next flight

Three changes follow directly from sections 5 to 7.

1. Record the reaction wheel and give it a chance to work. The wheel is the one subsystem I cannot grade, because nothing logged it. I will log its telemetry to SD, change the fault handling as listed in 7h, and add a swivel between the flight train and the payload. My hypothesis, which needs a flight test, is that a swivel matters more than a bigger wheel:

if it decouples the balloon's twist, the remaining disturbances may be small enough for ±3 deg/s of authority; without it, the calculation says about 2.4 times the band is needed.

2. Close the GPS gap. One receiver lost its fix for 52.3 min on ascent and 18 min on descent, and the Mega's own GPS columns were empty. I will fly two receivers, feed the second one to the Mega's Serial2 so the gps_* columns are populated, and log satellite count and fix quality every second. I will also rate the ascent-rate assumption in the prediction honestly: both flights rose slower than the model, 3.42 m/s on Flight 2 and about 1.62 m/s at 4 to 10 km on Flight 1. The slower ascent likely contributed to the 37.2 km landing error, but I cannot separate its effect from errors in the wind forecast or burst altitude.

3. Fix the sensors and cameras that failed. Shield the exterior temperature sensors and give them forced airflow so the stratosphere readings mean something. Replace the humidity sensors with one rated for low pressure, or drop humidity. Keep the BMP390 for the troposphere only. Check every mux channel with the flight box closed, so a repeat of the MCP9808 #2 failure shows up on the bench. Insulate and tie down all three cameras after a cold-chamber test. Two of three cameras stopping at two minutes and the third missing burst is the single biggest loss of this flight.



Figure 21 Mobius frame shortly after liftoff, camera clock 12:58:02 (about 19:42:55 UTC). The launch site, car, and crew are below.



Figure 22 Mobius frame at camera clock 14:35:28, about 21:20 UTC and about 19.8 km pressure altitude. The lens is wide-angle; the horizon's curve is mostly lens distortion.



Figure 23 Flight 2 payload where it landed, about 32.5 km (20.2 mi) east of Baker, CA: upright and intact, bridle lines across the lid.



Figure 24 Second view of the Flight 2 landing site, with the parachute in the foreground.

Appendix A Flight 1 notes

Flight 1 on Aug 7, 2026 carried only an Adafruit Ultimate GPS logging to its internal LOCUS memory, a SPOT Trace, and an action camera. I decoded 930 fixes at 15 s intervals in 3 logger sessions. The peak was 31,761 m (104,203 ft) at 23:57:05 UTC and the last fix is at 34.8328 N, 118.6526 W, 947 m, at 01:18:51 UTC on Aug 8. The launch time cannot be computed from the log: the last ground fix is at 19:39:40 UTC and the first airborne fix is 31 min later at 4,221 m, and a second gap of 132 min hides the climb from 10 to 29.5 km. The gaps line up with the logger's session boundaries. The straight-line distance is 158.9 km (98.7 mi); the distance along the recorded fixes is 171.8 km, a lower bound because of the gaps. The ascent was slow, about 1.62 m/s between 4 and 10 km and 1.1 m/s over the last 2 km before burst.

Appendix B Parts list

Subsystem	Parts
Structure	XPS foam box, reflective insulation wrap, black tape; Kaymont TX1200 balloon (Flight 2); parachute (model not recorded); lines (type not recorded)
Logger	Arduino Mega 2560 R3; Arduino Uno with Adafruit Ultimate GPS Logger Shield (SD card); PCA9548 I2C multiplexer; STEMMA QT cables
Sensors	MCP9808 ×3, TMP119, MS8607, BMP390, AHT20, VEML7700, LTR390, LIS3DH
Tracking	SPOT Trace satellite tracker; Adafruit Ultimate GPS (LOCUS logging, Flight 1)
Cameras	Mobius Maxi 4K; RunCam Thumb 2 ×2 (Flight 2); action camera (Flight 1)
Reaction wheel	Arduino Nano; Adafruit BNO085 IMU; Pololu 25D 20.4:1 gearmotor with encoder; Pololu G2 18v17 motor driver; custom 4-spoke wheel
Power	Nitecore USB power banks ×2 (one powers the reaction-wheel Nano); LiPo battery for the wheel motor; two battery holders

Appendix C File and script index

File	What it is
data/flight2_2026-10-28/SCI_000.CSV	Flight 2 science log: 12,245 S rows, 24,822 raw NMEA G rows, boot/detect/event rows
data/flight2_2026-10-28/Second_Successful_Launch.kml	Google Earth project: observed track split at gaps, 13 prediction runs, SPOT fixes, special-use airspace overlay
data/flight1_2026-08-07/Aug_7_2026_GPS_Fixes.csv	Flight 1 GPS: 930 LOCUS fixes
data/flight1_2026-08-07/First_Successful_Launch.kml	Flight 1 decoded track; red connectors mark gaps
firmware/HAB_Mega_Science/, firmware/HAB_Uno_Logger/	Flight 2 Mega (1.2.0-I2C) and Uno (SCI-UNO-1.3.0-GPS-UNO) sketches with sci_link.h
firmware/HAB_RW_Controller/	Reaction-wheel firmware HAB-RW 1.2.0-ground, its telemetry format, and flight_run_summary.txt (ground test)
rotation_results/	Rotation-script output: Flight_2_rotation.csv, Flight_2_upper_rotation.csv, per-window plots
analysis/common.py	Loaders for every raw file; computes every flight number quoted in this report (compute_numbers)
analysis/flight2_numbers.py	Prints the numbers as JSON
analysis/make_figures.py	The drawn figures (block diagrams, charts, maps) as 300-dpi PNGs in docs/figures/
analysis/make_altitude_csv.py	docs/flight2_gps_altitude_1min.csv: GPS altitude at one-minute intervals from release
analysis/build_report.py	Builds this PDF from the numbers and figures
docs/figures/	Every figure as a separate 300-dpi PNG

Regulatory note: I notified the FAA before each flight. This report makes no claim about FAR Part 101 compliance or exemption.

Not in the repository: the original video clips, the Flight 1 LOCUS binary, the wheel's raw telemetry and EEPROM bytes, CAD and bifilar-pendulum records, motor bench logs, the archived flight binaries, and the post-landing files SCI_001 to SCI_006. Claims that depend on them are labeled as observations or estimates.